MATH
数模加油站

美赛各版块高分写作套路

史上最全数学建模综合教程（数学建模写作、算法、编程从入门、速成到进阶）

数模写作官方规则|推荐结构|各版块技巧

主讲人：江北

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写作套路 | 官方书写准则

➢ **通用框架：**美赛官方对论文框架没有要求。但有准则如下（节选自**官网**）。

1、摘要： 评委们非常重视摘要，获奖论文通常会根据摘要的质量与其他论文区分开来。

- 要写出好的摘要，请想象读者会根据您的摘要选择是否阅读论文正文：您在摘要中的简洁陈述应该激发读者了解您工作的细节。
- 您应该最后写总结，因为它应该清楚地描述您解决问题的方法，并且最突出的是您最重要的结论。确保在解决问题后计划时间写出全面而清晰的总结。
- 仅仅重述竞赛问题或从引言中剪切粘贴样板的摘要通常被认为是薄弱的。

2、总体而言： 团队的解决方案应该清晰、简洁、有条理，以便读者能够轻松理解解决方案的过程和结论。关键陈述应提出主要想法和结果。

- 目录可帮助读者预览报告的组织结构。
- 酌情对问题进行澄清或重述。
- 对所有变量和假设进行清晰的阐述。
- 陈述并证明与问题相关的合理假设。
- 提出问题分析，激发或证明所使用模型的合理性。
- 在解决方案的正文中总结推导、计算或说明性示例，并在适当的附录中留下冗长的推导和/或计算和数据。
- 包括模型的设计。讨论如何测试模型，包括误差分析、灵敏度和/或稳定性。
- 讨论您的模型或方法的任何明显的优点或缺点。
- 明确提供结论并报告结果。
- 文档资源和参考文献。

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写作套路 | 推荐论文结构

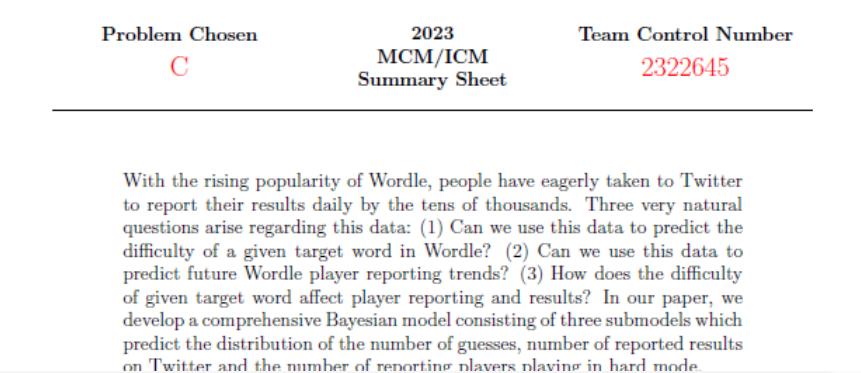
Title	
Summary	
Contents	
1 Introduction	1.1 Background
	1.2 Restatement
	1.3 Literature review
	1.4 Our work
2 Assumptions and Justifications	
3 Notations	
4 Model I: XXX	4.1 data preprocessing
	4.2 XXX model
	4.3 results of XXX
5 Model II: XXX	
6 Sensitivity Analysis	
7 Model Evaluation and Further Discussion	
9 Conclusion	
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Appendices	

标题	
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	1.2 问题重述
	1.3 现有文献综述
	1.4 工作思路
2 模型假设和合理性验证	
3 符号说明	
4 模型一: XXX	4.1 数据预处理
	4.2 XXX 模型
	4.3 求解结果
5 模型二: XXX	
6 灵敏度检验	
7 模型的评价及推广	
9 结论	
参考文献	
附录	

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写作套路｜1）论文标题/关键词怎么写（Title）

- **标题：美赛对于标题和关键词没有这么看重（部分O奖论文无标题和关键词）**
  - **两段式**：Crack the Wordle Puzzle: Word Attribute Analysis Approaches（破解 Wordle 之谜：词语属性分析方法）
  - **常规的：** Viral Spread Characteristics and Difficulty Determinants of Wordle Modeling Based on Differential Equations and K-nearest Neighbors（**基于**……**方法**的……研究）
  - **有吸引力的：** Uncover the Hidden Secrets in Wordle Results（揭开 Wordle 结果中隐藏的秘密）
  - **简约的：** Winners in Wordle（Wordle 中的获奖者）
  - **没有标题的**（建议还是写上标题）：



Problem Chosen: C

2023 MCM/ICM Summary Sheet

Team Control Number: 2322645

With the rising popularity of Wordle, people have eagerly taken to Twitter to report their results daily by the tens of thousands. Three very natural questions arise regarding this data: (1) Can we use this data to predict the difficulty of a given target word in Wordle? (2) Can we use this data to predict future Wordle player reporting trends? (3) How does the difficulty of given target word affect player reporting and results? In our paper, we develop a comprehensive Bayesian model consisting of three submodels which predict the distribution of the number of guesses, number of reported results on Twitter and the number of reporting players playing in hard mode.

标题介词冠词等（on, via, and, of, the）单词**小写**，其他单词首字母大写

- **关键词：**关键词一般3-6个，主要为题目关键词、使用的主要模型、算法名称、数学名词等，也有的论文干脆不写关键词（不建议）。

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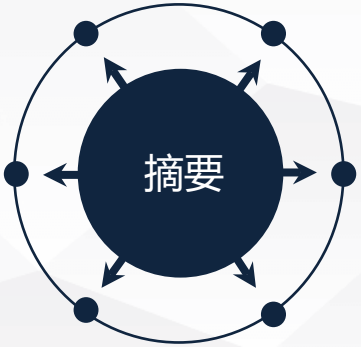
写作套路｜2）摘要怎么写（Summary）

摘要是重中之重！

评委们非常重视摘要，获奖论文通常会根据摘要的质量与其他论文区分开来（官网描述）。
总之，让评委愿意读下去，模型算法的专有名词不能错，每一小问，都要给一个明确的结果（数值或文字描述），每一个模型，都要结合题目去描述如何设置参数和求解，若摘要写的逻辑混乱、评委找不到模型和每一问的求解结果，就与获奖无缘。

基本要求

- 开头段整体描述，中间几段与正文的架构相同，结尾段自由发挥也可不写
- 中间几段的每一段从4方面描述：建立的模型+求解的过程/算法+得到的数值+对应的意义
- 摘要中对赛题的每一小问都有明确的结果
- 不要添加表格、图片



常用模板

开头段：针对什么问题，建立了什么模型

针对问题一，建立模型，模型细节，求解结果

针对问题二，……

针对问题三，……

结尾总结段，模型的优缺点，灵敏度分析等，以及 可以改进的方面（非必要）

关键词Keywords：模型名称，算法名称，问题关 键词等

注意摘要页控制在一页内

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写作套路｜2）摘要怎么写（Summary）

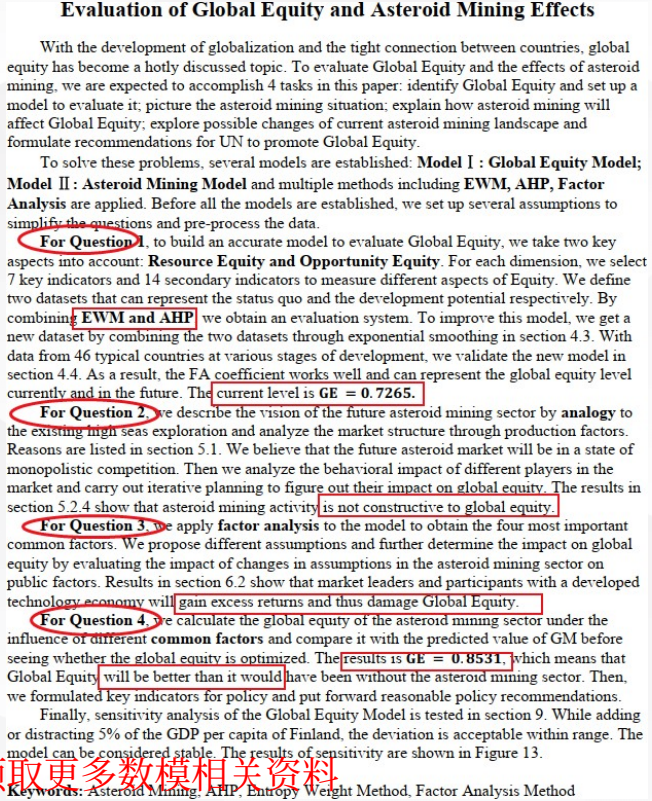
开头 | 中间段 | 总结 | 关键词

➢ 开头段： 介绍问题和所建立的模型

- With the development of XX, the XX problem has become a hot topic in the paper. In order to solve the XX problem, this paper establishes/utilizes the XX model and solves to obtain XX.

- 随着XX的发展，XX问题已经成文一个热点话题。为了解决XX问题，本文建立/利用了XX模型，求解得到XX.

- 第一段不必太长，4到6行即可。右图示例开头用了两段，略长。



Evaluation of Global Equity and Asteroid Mining Effects

With the development of globalization and the tight connection between countries, global equity has become a hotly discussed topic. To evaluate Global Equity and the effects of asteroid mining, we are expected to accomplish 4 tasks in this paper: identify Global Equity and set up a model to evaluate it; picture the asteroid mining situation; explain how asteroid mining will affect Global Equity; explore possible changes of current asteroid mining landscape and formulate recommendations for UN to promote Global Equity.

To solve these problems, several models are established: **Model Ⅰ: Global Equity Model; Model Ⅱ: Asteroid Mining Model** and multiple methods including **EWM, AHP, Factor Analysis** are applied. Before all the models are established, we set up several assumptions to simplify the questions and pre-process the data.

For Question 1, to build an accurate model to evaluate Global Equity, we take two key aspects into account: **Resource Equity and Opportunity Equity**. For each dimension, we select 7 key indicators and 14 secondary indicators to measure different aspects of Equity. We define two datasets that can represent the status quo and the development potential respectively. By combining **EWM and AHP** we obtain an evaluation system. To improve this model, we get a new dataset by combining the two datasets through exponential smoothing in section 4.3. With data from 46 typical countries at various stages of development, we validate the new model in section 4.4. As a result, the FA coefficient works well and can represent the global equity level currently and in the future. The current level is **GE = 0.7265.**

For Question 2, we describe the vision of the future asteroid mining sector by **analogy** to the existing high seas exploration and analyze the market structure through production factors. Reasons are listed in section 5.1. We believe that the future asteroid market will be in a state of monopolistic competition. Then we analyze the behavioral impact of different players in the market and carry out iterative planning to figure out their impact on global equity. The results in section 5.2.4 show that asteroid mining activity is not constructive to global equity.

For Question 3, we apply **factor analysis** to the model to obtain the four most important common factors. We propose different assumptions and further determine the impact on global equity by evaluating the impact of changes in assumptions in the asteroid mining sector on public factors. Results in section 6.2 show that market leaders and participants with a developed technology economy will gain excess returns and thus damage Global Equity.

For Question 4, we calculate the global equity of the asteroid mining sector under the influence of different **common factors** and compare it with the predicted value of GM before seeing whether the global equity is optimized. The results is **GE = 0.8531**, which means that Global Equity will be better than it would have been without the asteroid mining sector. Then, we formulated key indicators for policy and put forward reasonable policy recommendations.

Finally, sensitivity analysis of the Global Equity Model is tested in section 9. While adding or distracting 5% of the GDP per capita of Finland, the deviation is acceptable within range. The model can be considered stable. The results of sensitivity are shown in Figure 13.

Keywords: Asteroid Mining, AHP, Entropy Weight Method, Factor Analysis Method

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写作套路｜2）摘要怎么写（Summary）

开头 | 中间段 | 总结 | 关键词

➢ **中间段：**

需要对每个问题分别进行叙述，包含应用了什么方法；得到了什么结果；解决了什么问题。

- For question Ⅰ, For question Ⅱ, For question Ⅲ
- First, Second, Third
- 针对问题一，针对问题二，针对问题三
- 首先，其次，然后

- 每段描述清楚一小问的求解过程：建立的模型，模型的关键参数，求解的过程与结果，结果的意义
- 关键参数比如模拟退火降温系数、ARIMA的dpq等，评委看到对模型参数值设置和理由的描述，就明白你是真得懂这个模型或算法！印象分会好很多
- 若求解的对象比较少，可直接写数值，较多的话说明结果在哪个表格/图片即可

Evaluation of Global Equity and Asteroid Mining Effects

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写作套路｜2）摘要怎么写（Summary）

开头 | 中间段 | 总结 | 关键词

- **总结段：总结或描述灵敏度分析、优缺点等（非必要）**

  - Finally, our model……
  - 最后，我们的模型较好地解决了YY问题，模型在XX情况下能够准确求解YY，而在XX情况下需要考虑……。

  - 给自己一句低调的夸赞，再提一提注意事项，哪些条件下模型较好、哪些条件又不好（优缺点分析）以及灵敏度分析等。
  - 最后一段不是必要的，时间不够可以不写

- **关键词：问题背景关键词，建立的模型名称，算法名称等，4到6个左右即可**

Evaluation of Global Equity and Asteroid Mining Effects

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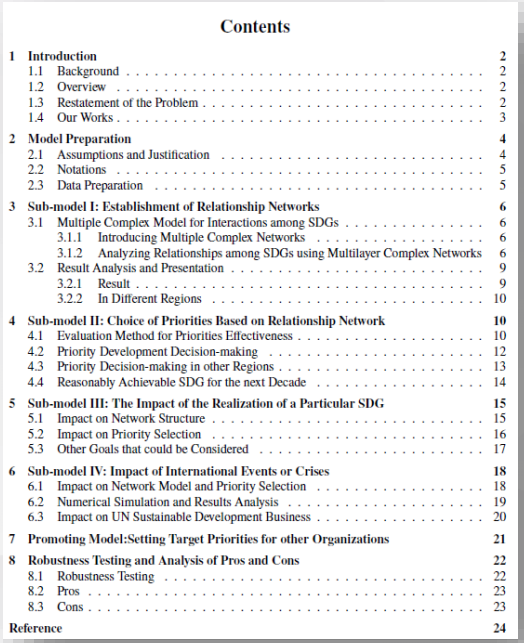
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写作套路｜3）目录（Contents）

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Contents就是**目录**，目录可以方便读者查看文章的整体框架。美赛的论文官方推荐放目录，华为杯研赛论文也要放目录。

- 注意不要手动写目录！！！Word/Latex本身有生成目录并更新的功能
  - 需要结合多级标题来使用
  - 目录需要利用分页符单独成页，
  - 后续会提供已设计好目录的论文模板

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写作套路｜4）引言（Introduction）

- **引言（Introduction）**：一般写background（背景）和Restatement of the Problem（问题重述）即可
  - 引言不是重点，不要花费太多时间；可以写完正文和摘要再写
  - 除了背景和问题重述，还可写Literature Review（文献综述）和工作思路（Our Work）

Background

Restatement of the Problem

Literature Review

Our Work

1. Background（背景）

简单来说，就是对赛题进行补充说明。

例如题目是关于新能源汽车的问题，题目中没有提到电池续航，所以可以查相关文献，用自己的话复述下文献里的一些语句即可。

建议：写完论文正文部分之后，再根据正文里的内容来写背景，但注意不要在这部分提到任何模型或解题思路，也不要照搬比赛题目中的内容。

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写作套路｜4）引言（Introduction）

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Background

Restatement of the Problem

Literature Review

Our Work

2. Restatement of the Problem（问题重述）

用自己的话，把赛题每一小问的关键语句重述一遍。该部分内容不要写的过多，专有名词不要更换

通用的模版：

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- *Problem 1*
- *Problem 2*
- *Problem 3*

注意，这个小部分也可以和问题背景合并到一起，作 为第一个小部分(和国赛类似)，大家根据自己的需要调整

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写作套路｜4）引言（Introduction）

- **引言（Introduction）**：一般写background（背景）和Restatement of the Problem（问题重述）即可
  - 引言不是重点，不要花费太多时间；可以写完正文和摘要再写
  - 除了背景和问题重述，还可写Literature Review（文献综述）和工作思路（Our Work）

Background

Restatement of the Problem

Literature Review

Our Work

3. Literature Review（文献综述）

文献综述就是把关于当前问题的现有研究成果做个概述。首先需要阅读大量解决该问题的论文，其次得用自己的话总结出来。

除非想冲O奖，否则别写这部分。一来竞赛时间有限，不可能去阅读大量论文；二来能力有限，不一定能写好总结。

小技巧：去搜相关论文，一般发表的论文都会有文献综述部分，照着别人的综述用自己的话描述一遍即可。

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写作套路｜4）引言（Introduction）

- **引言（Introduction）**：一般写background（背景）和Restatement of the Problem（问题重述）即可
  - 引言不是重点，不要花费太多时间；可以写完正文和摘要再写
  - 除了背景和问题重述，还可写Literature Review（文献综述）和工作思路（Our Work）

Background

Restatement of the Problem

Literature Review

Our Work

4. Our Work（工作思路）

这个小部分主要介绍论文的分析思路和建模的框架，有点像国赛论文中的问题分析部分，可以画个思维导图（可用软件亿图图示），大家可以学习。

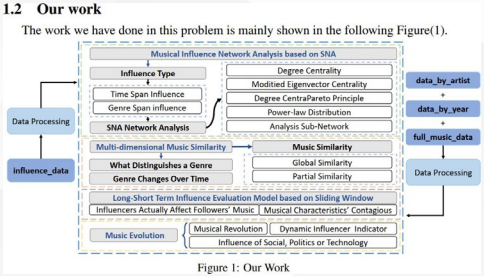


Figure 1: Our Work

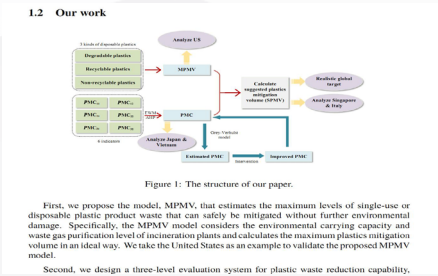


Figure 1: The structure of our paper.

First, we propose the model, MPMV, that estimates the maximum levels of single-use or disposable plastic product waste that can safely be mitigated without further environmental damage. Specifically, the MPMV model considers the environmental carrying capacity and waste gas purification level of incineration plants and calculates the maximum plastics mitigation volume in an ideal way. We take the United States as an example to validate the proposed MPMV model.

Second, we design a three-level evaluation system for plastic waste reduction capability,

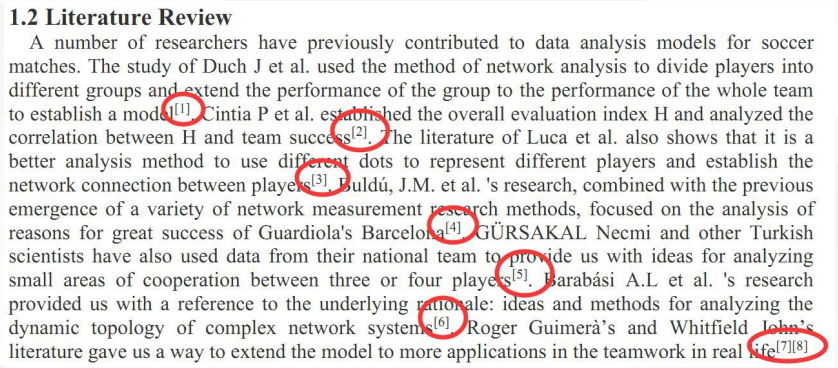
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写作套路｜4）引言（Introduction）

➢ 文献综述

在对题目所涉及的研究领域的文献进行广泛阅读和理解的基础上，对该研究领域的研究现状（包括主要学术观点、前人研究成果和研究水平、争论焦点、 存在的问题及可能的原因等）、新水平、新动态、新技术和新发现、发展前景等 内容进行综合分析、归纳整理和评论，并提出自己的见解和研究思路而写成的一 种文体。它要求作者既要对所查阅资料的主要观点进行综合整理、陈述，还要根 据自己的理解和认识，对综合整理后的文献进行比较专门的、全面的、深入的、 系统的论述和相应的评价，而不仅仅是相关领域学术研究的“堆砌”。

注意：文献综述本身就是一种文体，可以作为论文独立发表，也可以放在论 文中作为独立的一个部分；我们在美赛中写的文献综述不需要达到上面定义写的 这么严格的标准。



1.2 Literature Review

A number of researchers have previously contributed to data analysis models for soccer matches. The study of Duch J et al. used the method of network analysis to divide players into different groups and extend the performance of the group to the performance of the whole team to establish a model[1]. Cintia P et al. established the overall evaluation index H and analyzed the correlation between H and team success[2]. The literature of Luca et al. also shows that it is a better analysis method to use different dots to represent different players and establish the network connection between players[3]. Buldú, J.M. et al. 's research, combined with the previous emergence of a variety of network measurement research methods, focused on the analysis of reasons for great success of Guardiola's Barcelona[4]. GÜRSAKAL Necmi and other Turkish scientists have also used data from their national team to provide us with ideas for analyzing small areas of cooperation between three or four players[5]. Barabási A.L et al. 's research provided us with a reference to the underlying rationale: ideas and methods for analyzing the dynamic topology of complex network systems[6]. Roger Guimerà's and Whitfield John's literature gave us a way to extend the model to more applications in the teamwork in real life[7][8].

总结写作特点：

有很多学者都研究过这个问题(红色圈出来的编号就是参考文献的标记)，其中：

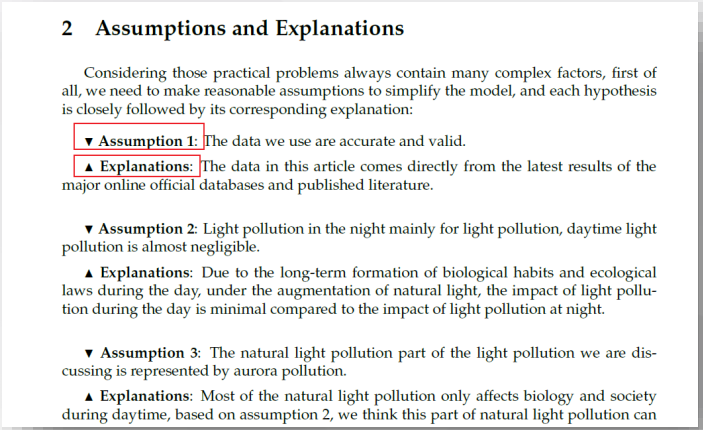
- 某某学者建立了某某模型或者使用了某某方法研究了这个问题；
- 某某学者通过研究了这个问题得到了某某结论。

技巧：文献综述中的文章能直接和你的论文中使用的模型或者得到的结论相互呼应！

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写作套路｜5）模型假设与说明（Assumptions and Justification）

➢ **模型假设与说明：**Justification翻译过来是正当理由的意思。这一部分要写模型假设，并且要对论证假设的合理性，这一点比国赛的要求要高，请大家引起足够的注意。



2 Assumptions and Explanations

Considering those practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

▼ **Assumption 1**: The data we use are accurate and valid.

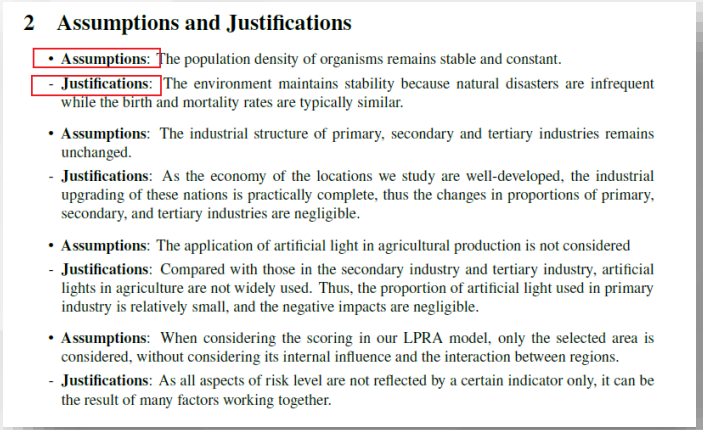
▲ **Explanations**: The data in this article comes directly from the latest results of the major online official databases and published literature.

▼ **Assumption 2**: Light pollution in the night mainly for light pollution, daytime light pollution is almost negligible.

▲ **Explanations**: Due to the long-term formation of biological habits and ecological laws during the day, under the augmentation of natural light, the impact of light pollution during the day is minimal compared to the impact of light pollution at night.

▼ **Assumption 3**: The natural light pollution part of the light pollution we are discussing is represented by aurora pollution.

▲ **Explanations**: Most of the natural light pollution only affects biology and society during daytime, based on assumption 2, we think this part of natural light pollution can



2 Assumptions and Justifications

- **Assumptions**: The population density of organisms remains stable and constant.
  - **Justifications**: The environment maintains stability because natural disasters are infrequent while the birth and mortality rates are typically similar.
- **Assumptions**: The industrial structure of primary, secondary and tertiary industries remains unchanged.
  - **Justifications**: As the economy of the locations we study are well-developed, the industrial upgrading of these nations is practically complete, thus the changes in proportions of primary, secondary, and tertiary industries are negligible.
- **Assumptions**: The application of artificial light in agricultural production is not considered
  - **Justifications**: Compared with those in the secondary industry and tertiary industry, artificial lights in agriculture are not widely used. Thus, the proportion of artificial light used in primary industry is relatively small, and the negative impacts are negligible.
- **Assumptions**: When considering the scoring in our LPRA model, only the selected area is considered, without considering its internal influence and the interaction between regions.
  - **Justifications**: As all aspects of risk level are not reflected by a certain indicator only, it can be the result of many factors working together.

□ 所有假设的目的都是为了在数学上简化问题

- 添加理想条件：遍历多个城市/旅行商类问题
- 假设：地图上的直线距离，视为两两城市之间的距离
- 现实中不同的交通方式（飞机、铁路、公路等）均可实现两两城市互通，选取最简单且理论上可行的情况，即直线距离，在数学上就转化为了图论问题。

- 忽略掉非重要因素：某篇2021年美赛C题特等奖论文的假设
- 假设：黄蜂活动时间是4月到10月，其他时间黄蜂在冬眠，发现的黄蜂事件可以忽略
- 当然现实中不可能一天不早、一天不晚真的严格按照4月1日到10月31日，而比赛期间无法得到准确数据，因此需要做出假设，在数学上规定4月到10月这段时间作为时间区间左右端点

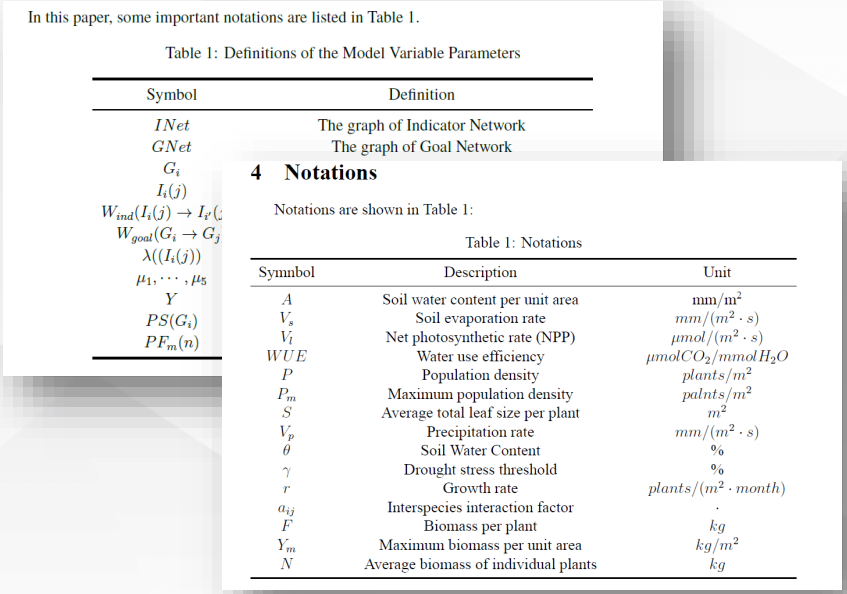
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写作套路 | 6）符号说明（ Notations ）

Notations是对模型中使用的重要变量进行说明，表格形式三线表，表头分别是Symbol（符号）、 Description （含义） 、 Unit （单位）（可不写），一般排版时尽量放到一页中。

注意：

- 只需写主要模型用到的重点变量、全篇通用的变量
- 求解计算等过程中的局部变量不要写
- 符号要以公式的形式写；如果是物理量，可在描述里写单位
- 每个符号的描述要简短，控制在一行内



In this paper, some important notations are listed in Table 1.

Table 1: Definitions of the Model Variable Parameters

Symbol	Definition
$INet$	The graph of Indicator Network
$GNet$	The graph of Goal Network
G_i	
$I_i(j)$	
$W_{ind}(I_i(j) \rightarrow I_{i'}($	
$W_{goal}(G_i \rightarrow G_j$	
$\lambda((I_i(j))$	
$\mu_1, \cdots, \mu_5$	
Y	
$PS(G_i)$	
$PF_m(n)$	

4 Notations

Notations are shown in Table 1:

Table 1: Notations

Symnbol	Description	Unit
A	Soil water content per unit area	mm/m^2
V_s	Soil evaporation rate	$mm/(m^2 \cdot s)$
V_i	Net photosynthetic rate (NPP)	$\mu mol/(m^2 \cdot s)$
WUE	Water use efficiency	$\mu molCO_2/mmolH_2O$
P	Population density	$plants/m^2$
P_m	Maximum population density	$palnts/m^2$
S	Average total leaf size per plant	m^2
V_p	Precipitation rate	$mm/(m^2 \cdot s)$
θ	Soil Water Content	%
γ	Drought stress threshold	%
r	Growth rate	$plants/(m^2 \cdot month)$
a_{ij}	Interspecies interaction factor	-
F	Biomass per plant	kg
Y_m	Maximum biomass per unit area	kg/m^2
N	Average biomass of individual plants	kg

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写作套路 | 7）数据描述(Data Description)

Data Description翻译过来就是数据描述。注意，这部分内容不是必须的，大家根据自己的需要来进行添加。

如果自己收集了数据或者题目给了数据的话，可以先对数据进行一个简单的介绍，或者将数据可视化，然后再从图形中得到一些直观的结论。

相关的术语

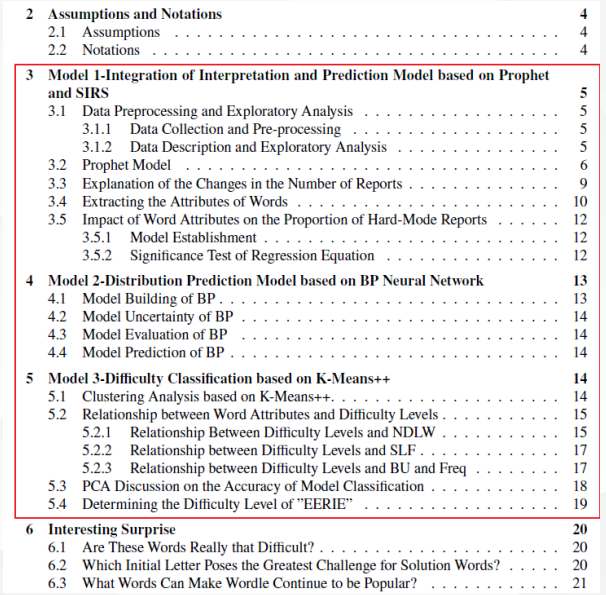
- Data Collection(数据的收集)
- Data Pre-processing(数据预处理)/Data Cleaning(数据清洗)
- Data Visualization(数据可视化)
- Descriptive Statistical Analysis of the Data(数据的描述性统计分析)

这部分内容在论文中的位置也比较灵活，我们可以将这个内容放到模型的建立与求解中，也有部分论文放在了引言部分，还有的论文将这个内容放到“Model Preparation模型准备”这个部分。特别的，如果做的是美赛C题（C题一般是数据分析类型的题目），我们可以把这个部分单独作为一个大的部分，然后进行数据预处理和数据可视化分析。

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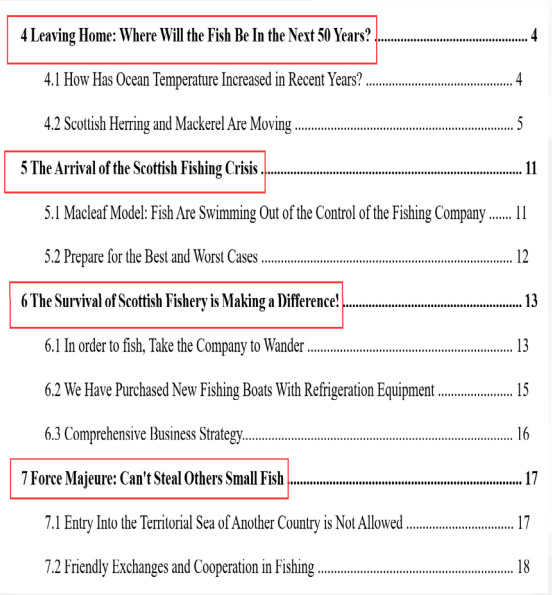
写作套路 | 8）正文结构

- 正文结构
  - 标题分级与命名
  - 标题最多3级（模板已设置好字体格式）
  - 命名格式多样（推荐以模型或小问来命名）
    1）以模型命名
    2）以具体问题命名
    3）以问题1，2，3命名
    4）一个标题命名
  - 这一部分内容一定要全面，包括必要的分析，建模过程，结果展示，结果分析等



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以模型命名（推荐）



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6.3 Comprehensive Business Strategy 16
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7.2 Friendly Exchanges and Cooperation in Fishing 18

以要解决的问题命名

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写作套路 | 8）正文结构

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模型放在一个大标题

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写作套路 | 9）灵敏度分析(Sensitivity Analysis)

灵敏度分析是研究与分析一个系统（或模型）的状态或输出变化对系统参数或周围条件变化的敏感程度的方法。通用步骤（控制变量法）：控制其它参数不变的情况下，改变模型中某个重要参数的值，然后观察模型的 结果的变化情况。

误差分析是指分析模型中的误差来源，或者估算模型中存在的误差，一般用于预测问题或者数值计算类问题。

灵敏度分析/误差分析



使用前检验/结果检验

模型的检验可以分为两种，一种是使用模型之前应该进行的检验，例如层次分析法中的一致性检验，灰色预测中的准指数规律的检验，这部分内容应当**放在模型的建立部分**；另一种是使用了模型后对模型的的结果的检验，最常见的稳定性检验（和灵敏度分析类似）。

Sensitivity Analysis翻译过来就是灵敏度分析或敏感性分析。

在美赛的写作中，写的最多的就是灵敏度分析，因此这里我们的标题就直接取得是Sensitivity Analysis.如果你既要写灵敏度分析，又要写误差分析，那么你可以把标题改成：Sensitivity Analysis and Error Analysis.

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写作套路｜10）模型的评价推广(Model Evaluation)

- **模型的评价与推广**（本部分的标题需要根据你的内容进行调整，如果你没有写进一步讨论的话，就直接把标题写成模型的评价Model Evaluation）

  - 所谓模型优缺点（**Strengths and Weaknesses**）评价并不局限于模型本身，在整个建模过程中所有的优缺点均可在最后进行陈述，一般撰写优缺点的基本原则是有点说充分，缺点不回避，但注意，**优点数量要多于缺点。**
  - 模型推广（**Further Discussion**）主要描述你的模型完成了什么工作？达到了什么目的？得出了什么规律？为今后的工作提供了什么思路？结果有什么理论或实际用途？

优点表述形式：

- ✓ 模型或思路设计的简介使用，效率高
- ✓ 本文模型具有很强的创新性
- ✓ 模型的计算结果准确，精度高
- ✓ 模型考虑的系统全面，有很强的实用性
- ✓ 对模型进行了各类检验、稳定性高
- ✓ 模型本身具有的一些优点

缺点表述形式：

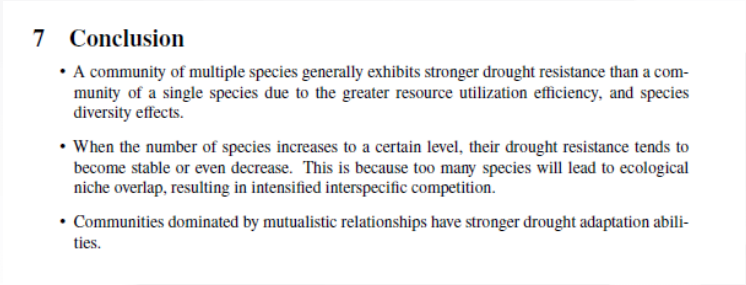
- ✓ 受某些因素限制，某些情况考虑
- ✓ 本文考虑的因素较为理想，降低了模型的普适度
- ✓ 由于系统考虑了某些因素，导致模型较为复杂，计算时间效率较低
- ✓ 模型本身的一些缺陷

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写作套路 | 11）结论(Conclusion)

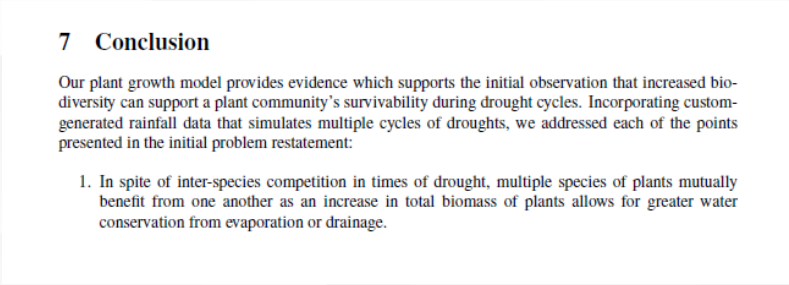
➢ 结论

- Conclusion翻译过来就是结论，这个部分在国赛论文几乎见不到，但在美赛中出现的频率很高。
- 这个部分可以是论文中心思想的重申、研究结果或主要观点的归纳，也可以是某些启示性的解释或考虑。
- 有些论文把“Model Evaluation and Further Discussion”的内容放到了结论部分，这也是可以的，大家可以灵活调整。



7 Conclusion

- A community of multiple species generally exhibits stronger drought resistance than a community of a single species due to the greater resource utilization efficiency, and species diversity effects.
- When the number of species increases to a certain level, their drought resistance tends to become stable or even decrease. This is because too many species will lead to ecological niche overlap, resulting in intensified interspecific competition.
- Communities dominated by mutualistic relationships have stronger drought adaptation abilities.



7 Conclusion

Our plant growth model provides evidence which supports the initial observation that increased biodiversity can support a plant community's survivability during drought cycles. Incorporating custom-generated rainfall data that simulates multiple cycles of droughts, we addressed each of the points presented in the initial problem restatement:

1. In spite of inter-species competition in times of drought, multiple species of plants mutually benefit from one another as an increase in total biomass of plants allows for greater water conservation from evaporation or drainage.

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写作套路 | 11）结论(Conclusion)

Summary(摘要)	摘要是对论文内容的一份简短陈述，需充分概括论文内容：解决了什么问题；应用了什么方法；得到了什么结果。 由于摘要不能超过一页，包含的内容也较多，因此摘要中的语句必须十分简练。 对于正文中使用的方法或建立的模型，摘要中不需要交代太多的细节，我们只需要清楚简明的概括出来即可；另外摘要中也不可能放上我们得到的所有结论，我们只需要放上最重要的结论。
Our Work(工作思路)	这个部分一般放在引言中，有点像国赛论文中的问题分析部分。我们可以结合题目的条件来对介绍我们的分析思路一般我们画一张漂亮的图形来展示整个建模的框架。 在这部分中，对于应用了什么方法要比摘要里面介绍的更加具体一点，但并不需要下任何结论。
Conclusion(结论)	在模型的建立和求解部分，我们得到了很多结论，我们可以将这些结论整理在本部分。 这部分只需要简单提一提我们用的方法，重点在于结论的总结和归纳。

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写作套路｜12）参考文献(References)

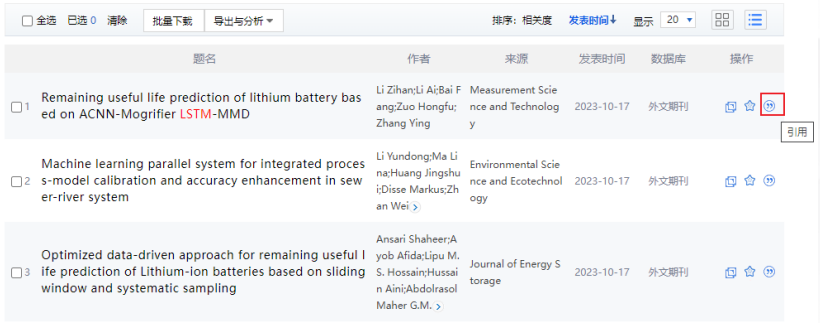
MATH 数模加油站

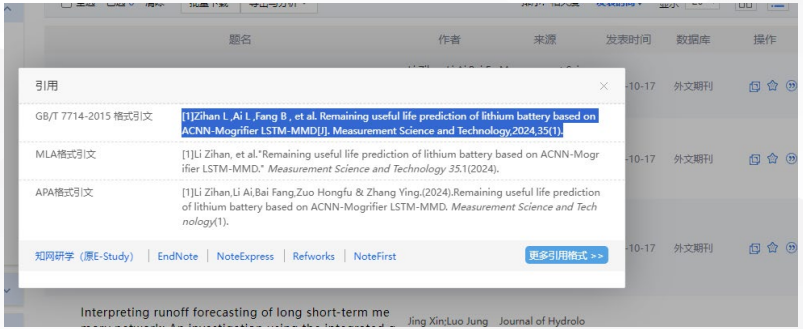
- **参考文献（References）：特别要注意格式正确**
  - 参考文献至少五六篇，引用中文文献记得翻译成英文。
  - 在正文对应部分也设置序号
  - 引用格式一定要正确，建议从检索网站直接导出，下图以知网为例

4.1.2 Population Predictions Using Modified Models

Combined with data from the National Bureau of Statistics[1] and related literature[2], we make population predictions for four plant species in the southeastern Tengger Desert of China. They are Salsola passerina, Reaumuria soongarica, Ceratoides latens and Kalidium foliatum. For convenience, we'd like to call them plant A, B, C and D.

The average monthly precipitation for the region from 2018 to 2021 is shown in Figure 1. We will predict and analyze the changes of the number of four plants in the 48 months through the modified model.





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写作套路 | 13）附录(Appendices)

- **附录：可以放入重要的代码、一些中间计算过程、复杂的推导等内容**
  - 可有可无。比赛规定**整个论文不能超过25页（包括附录，但不包括人工智能使用报告）**，所以完全可以不写附录
  - 写的话，选重要的代码放在表格里，写清简介



Appendices

Source Code on Competitive Model

```
import numpy as np
import matplotlib.pyplot as plt
from math import exp
import pandas as pd
import Get_hyphal_rate
import datapreocess

fungi_names = ['p.flav.s', 'p.har.n', 'm.trem.n', 'p.sang.s', 'h.seti.n']
def lineartransform(x):  # turn moisture to water potential
    y = -0.2 * x / 10
    return format(y, '.2f')

def f(hyphal_rate, temperature, factor):  # get decomposition rate by interpolation
    delta = [2.633 - 1.5139 * i / 120 for i in range(121)]
    w = int((temperature - 10) / 0.1)+delta[1.519
    return factor * (hyphal_rate ** (0.44))+factor/100/122
    return y

def get_m(humidity, temperature, fungi_names):  #  get the growth-hyphal extension ratio
    water_potential = lineartransform(humidity)
    m, factor = Get_hyphal_rate.get_hyphal_rate(fungi_names, water_potential,
                                                temperature)
    # get hyphal_rate: get hyphal extension rate under certain moisture and temperature
    return m, factor


delta_t, a, t_step = 1, 5, 122
F_0, l = np.ones((a, t_step)), np.ones((a, t_step-1)), np.ones((a, t_step))
V = np.zeros(t_step)
V[0], R = 10, 300
C = np.array([[1, 20, 5, 15, 10],
[5 - np.array([150, 100, 100, 100, 100])
Y, H = datapreocess.get_longTH(0)
for i in range(t_step-1):
    humidity = H[i]
    temperature = T[i]
    m, factor = get_m(humidity, temperature, fungi_names)
    r = np.diag(m/3)
    C = [1-ni/np.sum(m) for ni in m],
    C = np.array([[c[0], c[0], c[0], c[0], c[0]],
                  [c[1], c[1], c[1], c[1], c[1]],
                  [c[2], c[2], c[2], c[2], c[2]],
                  [c[3], c[3], c[3], c[3], c[3]],
                  [c[4], c[4], c[4], c[4], c[4]]])
    vector = F[:, i]/a
    B = np.diag(np.diag(r+F[:, i])*B
    F[:, i+1] = F[:, i]+delta_t*(B@vector)+delta_t
    l[:, i] = 1 - F[:, i+1]/R*np.sum(F[:, i])/K)*m.T
    V[i+1] = V[i]*(1-np.sum(F[:, i])/K)+v_temperature, factor)
    V[i+1] = V[i]*(1-np.sum(F[:, i])+delta_t*V[i]*(1-np.sum(F[:, i]))/K)

t = [i+delta_t for i in range(t_step)]
plt.plot(t, F[0, :], c='steelblue', label='p.flav.s')
plt.plot(t, F[1, :], c='olivedrab', label='p.har.n')
plt.plot(t, F[2, :], c='forestgreen', label='m.trem.n')
plt.plot(t, F[3, :], c='firebrick', label='p.sang.s')
plt.plot(t, F[4, :], c='mediumpurple', label='h.seti.n')
plt.xlabel("Time(day)", fontsize=16)
plt.ylabel("The density of fungi*m**3*(g/m*+*\u00b2*+*)", fontsize=16)
plt.legend(fontsize=12)
plt.show()

plt.plot(t, V, c='b', alpha=0.6, linewidth=2)
plt.xlabel("Time(day)", fontsize=14)
plt.ylabel("Relative density of woody fibers", fontsize=14)
plt.tick_params(labelsize=12)
plt.show()
```

Source Code on Getting hyphal rate

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def get_name():
```



Appendices

Appendix 1 —— Tools and software

Lingo & Matlab

Appendix 2 —— Code

Appendix 2.1

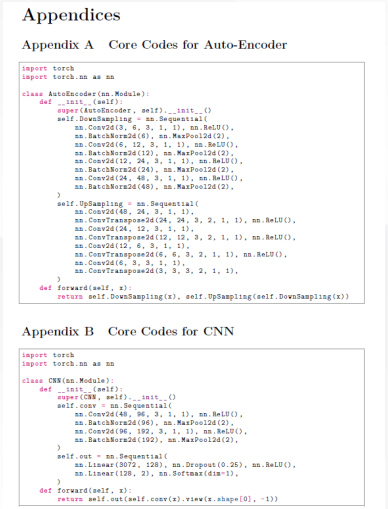
Introduce: Regional Discretization Model

```
sets:
x/1..COLS/:
y/1..ROWS/:
actions/1..5/:
link(x,y):H,totaltime;
link3(x,y,actions):t,delta;
endsets
dronenum = @sum(link3(i,j,k):delta(i,j,k));
value = @sum(x(i):@sum(y(j):(1-@exp(lambda*@sum(ac-
tions(k):t(i,j,k)))*H(i,j)) - boundary * dronenum;
@gin(ecox);
@gin(ecoy);
ecox=COLS;
ecoy=ROWS;
@for(link3(i,j,k):@bin(delta(i,j,k)));
@for(link(i,j):t(i,j,1)=delta(i,j,1)*(@max-@sqrt((ecox-i)^2+(ecoy-
j)^2)*dtime));
@for(link(i,j)|i&gt;1:t(i,j,2)=(delta(i,j,2)*(@max-@sqrt((ecox-i-0.5)^2+(ecoy-
j)^2)*dtime) + delta(i-1,j,2)*(@max-@sqrt((ecox-i+0.5)^2+(ecoy-
j)^2)*dtime))/2);
@for(link(i,j)|j&gt;1:t(i,j,3)=(delta(i,j,3)*(@max-@sqrt((ecox-i)^2+(ecoy-j-
0.5)^2)*dtime) + delta(i,j-1,3)*(@max-@sqrt((ecox-i)^2+(ecoy-
j+0.5)^2)*dtime))/2);
@for(link(i,j)|i&gt;1 #and# j&gt;1:t(i,j,4)=(delta(i-1,j,4)*(@max-@sqrt((ecox-
i+0.5)^2+(ecoy-j-0.5)^2)*dtime) +delta(i,j-1,4)*(@max-@sqrt((ecox-i-
0.5)^2+(ecoy-j+0.5)^2)*dtime))/2);
@for(link(i,j)|i&gt;1 #and# j&gt;1:t(i,j,5)=(delta(i,j,5)*(@max-@sqrt((ecox-i-
0.5)^2+(ecoy-j-0.5)^2)*dtime) +delta(i-1,j-1,5)*(@max-@sqrt((ecox-
i+0.5)^2+(ecoy-j+0.5)^2)*dtime))/2);
@for(link(i,j):H(i,j)=@if(i=1,j)*(ecox-1)*(COLS+1)-(ecoy-j))^2);
@for(link(i,j):totaltime(i,j)=@sum(link3(i,j,k):t(i,j,k)));
value = @sum(x(i):@sum(y(j):(1-@exp(lambda*@sum(ac-
tions(k):t(i,j,k)))*H(i,j))) - boundary * dronenum;
max =value;
```

Appendix 2.2

Introduce: Cellular automata simulation

```
a = 500;
b = 400;
k = 0.005;
veg = b*ones(a,b);
imb = image(c:\veg,veg,veg));
N = fspecial('gaussian',[81,81],0.9);
vegpast = zeros(a,b);
resultmat = zeros(5,25);
for burn = 0.6:0.02:1
    for turn = 1:5
```



Appendices

Appendix A Core Codes for Auto-Encoder

```
import torch
import torch.nn as nn

class AutoEncoder(nn.Module):
    def __init__(self):
        super(AutoEncoder, self).__init__()
        self.DownSampling = nn.Sequential(
            nn.Conv2d(3, 6, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(6), nn.MaxPool2d(2),
            nn.Conv2d(6, 12, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(12), nn.MaxPool2d(2),
            nn.Conv2d(12, 24, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(24), nn.MaxPool2d(2),
            nn.Conv2d(24, 48, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(48), nn.MaxPool2d(2),
        )
        self.UpSampling = nn.Sequential(
            nn.Conv2d(48, 24, 3, 1, 1),
            nn.ConvTranspose2d(24, 24, 3, 2, 1, 1), nn.ReLU(),
            nn.Conv2d(24, 12, 3, 1, 1),
            nn.ConvTranspose2d(12, 12, 3, 2, 1, 1), nn.ReLU(),
            nn.Conv2d(12, 6, 3, 1, 1),
            nn.ConvTranspose2d(6, 6, 3, 2, 1, 1), nn.ReLU(),
            nn.Conv2d(6, 3, 3, 1, 1),
            nn.ConvTranspose2d(3, 3, 3, 2, 1, 1),
        )
    def forward(self, x):
        return self.DownSampling(x), self.UpSampling(self.DownSampling(x))
```

Appendix B Core Codes for CNN

```
import torch
import torch.nn as nn

class CNN(nn.Module):
    def __init__(self):
        super(CNN, self).__init__()
        self.conv = nn.Sequential(
            nn.Conv2d(48, 96, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(96), nn.MaxPool2d(2),
            nn.Conv2d(96, 192, 3, 1, 1), nn.ReLU(),
            nn.BatchNorm2d(192), nn.MaxPool2d(2),
        )
        self.out = nn.Sequential(
            nn.Linear(3072, 128), nn.Dropout(0.25), nn.ReLU(),
            nn.Linear(128, 2), nn.Softmax(dim=1),
        )
    def forward(self, x):
        return self.out(self.conv(x).view(x.shape[0], -1))
```

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- **美赛有些题目会要求大家给某个杂志写一篇文章(Article)，例如：**

1）2020A题

In addition to your technical report, prepare a one- to two-page article for Hook Line and Sinker magazine to help fishermen understand the seriousness of the problem and how your proposed solution(s) will improve their future business prospects.

除了你的技术报告，准备一到两页的文章，作为 Hook Lineand Sinker的杂志内容，以帮助渔民了解问题的严重性，以及你的解决方案将如何改善他们未来的业务前景。

2）2020B题

Finally, write an informative, one- to two-page article describing your model and its results for publication in the vacation magazine: Fun in the Sun, whose readers are mainly non-technical.

最后，写一篇内容丰富的一到两页的文章，描述你的模型及其结果，发表在假期杂志上：FunintheSun，读者都是小白，不懂太复杂的模型。

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写作套路 | 14) Article，Letter and Memo

HOOK LINE AND SINKER 02/2020 Page1

Moving North!

Scottish Fishery Faces Huge Challenges



In recent years, the whole earth is warming and ocean temperatures have gradually increased. Herring and mackerel populations, who had previously lived comfortably off the coast of Scotland, felt this change around them, and they were forced to migrate to northern waters that were cooler than their current habitat.

Herring and mackerel represent a significant economic contribution to the Scottish fishing industry. Changes in population locations of herring and mackerel could make it economically impractical for smaller Scotland-based fishing companies.

Recently, a mathematical modeling team has studied the possible migration paths of these two fish species in the next 50 years. In their research report, they pointed out that Scottish fishery is in a very severe situation, and if no measures are taken in time, Scotland will have no fish to catch as soon as 13 years.

As the temperature of the ocean continues to rise, today's habitats for herring and mackerel are no longer suitable for their survival, and they will gradually move north to find habitats with a suitable temperature. According to the team's research, herring has now begun the process of migration and is expected to gradually migrate to Shetland within this year. In the future, herring populations will gradually move northwest, and they are expected to reach near Iceland by 2070.The team also pointed out that mackerel is constantly moving north in search of new habitat. Unlike herring, they may move northeast. It is expected that by 2050, mackerel populations may settle in the Norwegian sea.

As the global ocean temperature rises, the herring and mackerel are moving north faster. Many fishing companies in Scotland today are small-scale and do not have refrigeration equipment on the fishing vessels they use. This incident caught them off guard, as it almost made it impossible to catch distant fish and keep them fresh.

The fishing industry in Scotland comprises a significant proportion of the United Kingdom fishing industry. A recent inquiry by the Royal Society of Edinburgh found fishing to be of much greater social, economic and cultural importance to Scotland than it is relative to the rest of the UK. Scotland has just 8.4% of the UK population but lands at its ports over 60% of the total catch in the UK.

This shows how important fishing is to Scotland. The modeling team also clearly pointed out the fact that if the company continues to operate as it is today, then it will face a crisis of failure after many years.





Global Warming

Global warming is the long-term rise in the average temperature of the Earth's climate system. It is a major aspect of climate change and has been demonstrated by direct temperature measurements and by measurements of various effects of the warming. While there have been prehistoric periods of global warming, many observed changes since the mid-20th century have been unprecedented over decades to millennia.

Although the most common measure of global warming is the increase in the near-surface atmospheric temperature, over 90% of the additional energy stored in the climate system over the last 50 years has warmed ocean water. The remainder of the additional energy has melted ice and warmed the continents and the atmosphere.

The warming evident in the instrumental temperature record is consistent with a wide range of observations, documented by many independent scientific groups; for example, in most continental regions the frequency and intensity of heavy precipitation has increased. Further examples include sea level rise, widespread melting of snow and land ice, increased heat content of the oceans, increased humidity, and the earlier timing of spring events, such as the flowering of plants.



Page2 02/2020 HOOK LINE AND SINKER



Forward Fishery

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One might think this is a rather weird choice, but after further thinking, it soon turns out that mackerel and herring are the perfect objects for design because it is so important to Scotland. Yes, humble mackerel and herring are one of the pillars of our economy. Many of us have remembered the two kinds of fishes since childhood, and they were picked up from the end of the dock with a string of feathers hanging in buckets. They are provided as nutritious food resources and do make a significant contribution to Scotland's well-being. It is Scotland's most important catch until now. And they are considered as two kinds of the delicious fish around both in value and quantity.

Today, herring and mackerel as important totems may have left Scotland. Global warming is forcing them to leave here.

This could be a huge blow to Scotland. The modeling team who studied herring and mackerel migration paths is willing to help us solve this problem together. The team sent a representative as a special consultant, and he had an in-depth exchange with our North Atlantic Fisheries Management Association of Scotland. The team representative said that if the fishing company wants to survive, it must change! One of the suggestions they gave was to relocate the company, so that the company can get closer to the fish population that is gradually away.

The solution given by the modeling team is described as follows:

Solution A is a method that takes into account the possible migration path of fish in the next 50 years. Here we will plan only one new location for the company in 50 years (2020-2070). This will allow you to move again for 50 years no need to relocate. This can save a lot of money, and the problem of fishing has been solved. Of course, this may make the company's fishing range not reach the maximum utilization rate in some years, but it is also very reasonable to comprehensively consider the issue of funds.

Solution B is a method for a company that happens to be financially strong, that is, the company moves to a new location every 10 years. In this way, the maximum utilization of the fishing range in each time period can be achieved. The downside is that you need to invest a lot of money, but if the company can get enough of these investments, it is very good.

In addition, the team also provided us with proposals for new fishing vessels. These new fishing boats are much more advanced than the older ones because they have refrigeration equipment on them, which keeps the fish fresher on board.



Warming ocean waters threaten marine ecosystems and human livelihoods. For example, warm waters jeopardize the health of corals, and in turn, the communities of marine life that depend upon them for shelter and food. People who depend upon marine fisheries for food and jmay face negative impacts from the warming ocean.

The modeling team stated that the above schemes are predictions and suggestions based on existing conditions. They hope to discuss research with the people of Scotland and work out better strategies. Scottish fishing is bound to return to glory!



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➢ **美赛有些题目会要求大家写一封信(Letter)，例如：**

1）2020C题

Write a one- to two-page letter to the Marketing Director of Sunshine Company summarizing your team’s analysis and results. Include specific justification(s) for the result that your team most confidently recommends to the Marketing Director.

写一两页的信给阳光公司市场总监，总结您团队的分析和结果。包括针对您的团队 最有信心推荐给市场总监的结果的具体理由。

2）2019A题

Once your dragon analysis is complete, draft a two-page letter to the author of A Song of Ice and Fire, George R.R.Martin, to provide guidance about how to maintain the realistic ecological underpinning of the story, especially with respect to the movement of dragons from arid regions to temperate regions and to arctic regions.

当你完成对于龙的问题分析后，请写一封两页的信给“冰与火之歌”的作者George R.R.Martin，以提供关于如何保持故事的现实生态基础的指导，特别是关于从干旱地区 到温带地区和北极地区的龙。

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Solution A is a method that takes into account the possible migration path of fish in the next 50 years. Here we will plan only one new location for the company in 50 years (2020-2070). This will allow you to move again for 50 years no need to relocate. This can save a lot of money, and the problem of fishing has been solved. Of course, this may make the company's fishing range not reach the maximum utilization rate in some years, but it is also very reasonable to comprehensively consider the issue of funds.

Solution B is a method for a company that happens to be financially strong, that is, the company moves to a new location every 10 years. In this way, the maximum utilization of the fishing range in each time period can be achieved. The downside is that you need to invest a lot of money, but if the company can get enough of these investments, it is very good.

In addition, the team also provided us with proposals for new fishing vessels. These new fishing boats are much more advanced than the older ones because they have refrigeration equipment on them, which keeps the fish fresher on board.



Warming ocean waters threaten marine ecosystems and human livelihoods. For example, warm waters jeopardize the health of corals, and in turn, the communities of marine life that depend upon them for shelter and food. People who depend upon marine fisheries for food and jmay face negative impacts from the warming ocean.

The modeling team stated that the above schemes are predictions and suggestions based on existing conditions. They hope to discuss research with the people of Scotland and work out better strategies. Scottish fishing is bound to return to glory!



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写作套路 | 14）Article，Letter and Memo

➢ **美赛有些题目会要求大家写一封信(Letter)，例如：**

1）2020C题

Write a one- to two-page letter to the Marketing Director of Sunshine Company summarizing your team’s analysis and results. Include specific justification(s) for the result that your team most confidently recommends to the Marketing Director.

写一两页的信给阳光公司市场总监，总结您团队的分析和结果。包括针对您的团队 最有信心推荐给市场总监的结果的具体理由。

2）2019A题

Once your dragon analysis is complete, draft a two-page letter to the author of A Song of Ice and Fire, George R.R.Martin, to provide guidance about how to maintain the realistic ecological underpinning of the story, especially with respect to the movement of dragons from arid regions to temperate regions and to arctic regions.

当你完成对于龙的问题分析后，请写一封两页的信给“冰与火之歌”的作者George R.R.Martin，以提供关于如何保持故事的现实生态基础的指导，特别是关于从干旱地区 到温带地区和北极地区的龙。

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Letter

Dear Sunshine Company Marketing Director:

We are honored to inform you of our recommendations for product improvement and sales strategies for your company after data analysis and modeling. The following are some suggestions based on our analysis.

1. Suggestions for product improvement

We apply the LDA analysis model to find the topic feature of each product, in which the negative topic words can reflect the customers' dissatisfaction with the existing products in the market.

Analysis of the review text revealed that the words like "cost", "heavy", "smoking" and "noise" appeared on the top of the list of negative reviews on the hair dryer. After analyzing these high-frequency words one by one, we discovered that there were some drawbacks with hair dryers from the customers' point of view:

- Inefficiency: it takes a long time to blow-dry.
- Weight: it is too heavy to use conveniently.
- Bad quality: when using the product, there will be a lot of noise sometimes.

Therefore, according to the feedback above, we believe that the improvement of hair dryer products is supposed to start from improving the working efficiency and quality of hair dryer. Lightweight materials can be used in the production, reducing the weight of the hair dryer, should also reduce the noise when it works. For instance, lightweight materials can be used to reduce the weight of the hair dryer and reduce the noise while it is working.

As for microwave oven, words like "service" and "small" appeared on the top of the list of negative reviews. According to the feedback, we believe that the microwave oven may have the following defects:

- Size: someone considered that microwave ovens are too small, while others reckon that small ones are more convenient to use.
- Service: the after-sale service of some microwave ovens is not consummate.

In view of the above two points, we hold that microwave oven products can be designed in different sizes to meet the diverse needs of different customers. Moreover, improving service level and customer satisfaction is also a top priority.

"Hot" is the most frequently used negative word for a pacifier. Some customers think that the insulation function of the pacifier is poor, which reminds us that we should enhance the thermal insulation function in the design of the pacifier. In addition, we have noticed that customers attach great importance to the "appearance" of the pacifier and hence the cute product design will help to sell the product.

2. Sales strategy

Through the in-depth analysis of star ratings and reviews, we will propose sales suggestions and strategies from three aspects: the product brand, the time of selling products and the evaluation psychology of customers.

- Recommend the product brand for sale.

8 Letter to G.R.R.M.

January 28th, 2019

Dear Mr. George R. R. Martin,

We are writing to you today to discuss the ecological realism of the dragons in your popular book series Game of Thrones. We are a group of mathematicians and scientists who have meticulously considered the characteristics of your dragons and their sustainability on Earth. In the course of our analysis, we also explored the requirements of dragons, especially in regards to how they change in their environments. Although your series is fantasy and you likely did not impose the same rigorous realistic analysis that we did, we believe that the results of our exploration will nonetheless be of interest to you in your writing. After all, it can aid in creating the most realistic possible world in your writing.

In your series, you have three still-living dragons: Viserion, Rhaegar, and Drogon. To summarize our analysis, we must first discuss the known characteristics of the dragons we based it on, as per facts provided in your series. Firstly, we used the information that the dragons in your series are obligate carnivores, and thus only consume meat and fish. This was an important aspect of our mathematical model in determining the food resources required for each dragon's survival. Additionally, we based our mathematical exploration of the dragons' growth pattern on the fact that dragons in Westeros grow indefinitely, so long as they are provided with enough resources and space. As a result, we were able to compare your dragons to existing species with indeterminate growth to find their growth patterns.

Using these values, we were able to create an equation to estimate the amount of calories required for each dragon. Based on the sizes of your dragons, they would require different caloric intakes. However, using Balerion, the largest dragon recorded, we determined that the largest dragons would need at least 3.6 cows a day. As such, we recommend that you adhere to this food limit to maintain a realistic metabolic rate for your dragons.

With the help of the previously mentioned caloric requirement model, we were able to also determine the adjustments depending on climate. In your series, the three dragons commonly migrate to different settings, such as their time in the desert or their battle in the cold North. As a result, it is important to consider the effects of different environments on the dragons and adjust them accordingly to achieve maximal realism.

We looked at the effects of arid conditions on the dragons. Using data from existing migratory birds, we discovered that your dragons could likely produce water as a byproduct of their metabolism of their foods. As a result, in arid conditions, they would not require constant water provisions. However, there is a limit to this water amount which we recommend you adhere to for biological accuracy. Based on information from Season 5 Episode 5 of the show, we estimated that a dragon requires 49.46 kilograms of water per day. Based on their caloric intake, they would be able to produce this water solely from their metabolism for 7.5 hours of flight. We therefore recommend that in arid conditions, your dragons do not travel for longer than 7.5 hours without being provided with additional food or water. Due to the high food and water requirements of the dragons, while dragons can exist in arid conditions, it is not an optimal survival. Hence, we also recommend that in your series, dragons do not spend long amounts of time in such environments, since they could not feasibly fulfill their water requirements. In the event that arid conditions are necessary for plot purposes, we suggest that you take the high food and water requirements into account and provide the dragons with their necessary

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写作套路 | 14）Article，Letter and Memo

➢ **美赛有些题目会要求大家写一份备忘录(Memo)，例如：**

1）2020E题

Write a two-page memo to the ICM describing a realistic global target minimum achievable level of global single-use or disposable plastic product waste, a timeline to reach this level, and any circumstances that may accelerate or hinder the achievement of your target and timeline.

向 ICM 写一份两页的备忘录，描述全球一次性塑料生产废物达到全球最 小目标值的水平的时间表，并分析可能加速或阻碍该时间表的任何情况。

2）2019C题

In addition to your main report, include a 1-2 page memo to the Chief Administrator, DEA/NFLIS Database summarizing any significant insights or results you identified during this modeling effort.

除了主要报告之外，还要向DEA/NFLIS数据库的首席管理员提供1-2页的 备忘录，在备忘录中总结您在此建模过程中发现的任何重要见解或结果。

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8 Memo

To: International Council of Plastic Waste Management (ICM)

From: Team # 2017963

Date: 2020/2/16

Re: A realistic global target and its timeline

In response to your questions regarding global plastic waste mitigation, I'm writing to inform you of our findings.

We first propose the model, MPMV, that estimates the maximum levels of single-use or disposable plastic product waste that can safely be mitigated without further environmental damage. Then, we design a three-level evaluation system for plastic waste mitigation capacity, PMC, to investigate the minimal achievable level of global waste of single-use or disposable plastic products. Thereafter, we propose a target for the minimum achievable level of global waste of single-use or disposable plastic products in 2030 via perturbing the indicators in the PMC model, i.e., increasing the PMC curve to 0.6 by 2030. Finally, we put forward the SPMV model via combining MPMV and PMC models, to ensure the equity of task distribution of plastic waste reduction among countries.

Based on the in-depth analysis of the proposed models, we highlight the following insights:

- MPMV model investigates the mitigation of degradable, recyclable, and non-recyclable plastic product waste, and estimates the maximum levels of single-use or disposable plastic product waste that can safely be mitigated.
- PMC model considers the six indicators indicating generating less and mitigating more single-use or disposable plastic product waste, and finds that HDI is the most effective indicator to the levels of mitigation of plastic waste.
- The perturbed PMC model can be applied to estimate the target for the minimum achievable level of global waste of single-use or disposable plastic products, and it can be evaluated by predicting the PMC curve utilizing Grey-Verhulst model.
- It is a dynamic cycle process to establish the target for the minimum achievable level of plastic product waste in the PMC model. Therefore, the target can be dynamically adjusted according to the actual situations, and thereby it is possible to propose more effective countermeasures.
- Both MPMV model and PMC model ignore the equity of task distribution of plastic waste reduction among countries. Thus SPMV model combining MPMV and PMC models can more appropriately distribute the responsibility of mitigating global plastic waste according to the country's size and its current development status.

The proposed models refine the global target for the minimum achievable level of global single-use or disposable plastic product waste in 2030. That is, the degradable plastics will account for 10% of total plastic production, while the plastic recycling rate will be increased to 50%, and the ability to incinerate plastic product waste achieves the maximum. To achieve the target for the minimum achievable level of plastic product waste, the state, businesses, and consumers need to work together. First, ICM should advocate that global countries include

Memo

To: Chief Administrator, DEA/NFLIS Database
From: Team # 1909434
Date: 28 January 2019
Subject: Opioid Use Profile and Recommended Stategies in 5 States

Dear Chief Administrator, we are honored to inform you our summary and recommendation after data analysis and modeling.

Opioid Drug Use Profile

Here is the opioid use profile in Ohio, Pennsylvania, Virginia, West Virginia and Kentucky:

In general, heroin use increased rapidly from 2011 to 2015 and then decreased since 2015. Fentanyl use started to increase since about 2013 and the increasing speed was fairly drastic. Other opioids use didn't show drastic variations during 2010-2017. Thus, we infer that the fentanyl and its derivatives became a substitute for heroin. Quantitatively, without policy interventions, heroin use may continue to decrease and fentanyl use may continue to increase while other opioids use keep stable or decrease slowly in the future. Another notable find is that opioid drug use accounts for an increasing proportion of total opioid use.

Opioid abuse is a more serious problem in Ohio and Pennsylvania than other three states, and the number of drug reports is increasing rapidly in Ohio while decreasing in Pennsylvania. The variation trends of Kentucky, Virginia and West Virginia are relatively stable. The number of opioid reports in Kentucky almost equals that in Virginia although Virginia has nearly twice the population of Kentucky.

As we predict, the degree of heroin use will decrease at a relatively rapid speed in five states, and the total number of heroin report will be less than 10000 in 2019. However, fentanyl abuse is getting worse. The total number of fentanyl and its derivatives report will be more than 66000 in 2018 and 110000 in 2019. As for other opioid drugs, the use density is slowly increasing, but the degree of abuse of these drugs will not rise to a severe level in a short time.

Recommended Stategies

Our analysis also shows that opioid use is related to several socio-economic factors. Here we give our recommended strategies based on important socio-economic attributes identified.

- Strengthen the prohibition against addictive opioids like fentanyl(and its derivatives) in five states.
- Enforce road(to Virginia, West Virginia and Kentucky) transportation management and control by State Highway Patrol to contain the spread of opioid drugs.
- Turn attention to fighting against fentanyl and its derivatives compared with other drugs because fentanyl compounds are fairly potent and fentanyl use are drasticly increasing.
- Further popularize education for all citizens to reduce the number of school dropouts and strengthen drug addiction prevention and control education in campus.

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